

**PERFORMANCE WORKSTATEMENT
FOR
THE INSPECTION, MAINTENANCE AND
SERVICING OF THE
AIRCRAFT FIRE TRAINING FACILITY (AFTF)
AT
(Joint Base Andrews)**

(February 25, 2026)

**PERFORMANCE WORK STATEMENT
FOR
THE INSPECTION, MAINTENANCE AND SERVICING
OF THE
AIRCRAFT FIRE TRAINING FACILITY**

1. DESCRIPTION OF SERVICES. The contractor shall provide all tools, supplies, labor equipment and labor/management necessary to maintain and repair the Aircraft Fire Training Facility (AFTF) at Joint Base Andrews (JBA) in a manner that will ensure its serviceability and prevent deterioration in operational effectiveness. Component of the AFTF System are listed in Appendix A.

1.1. The contractor shall comply with manufacturer's recommendations and procedures, Air Force Technical Order (TO) 35E1-2-13-1 *Operation and Maintenance Instructions for Aircraft Fire Training Facilities*, 25 June 2013 Edition with Change 2 dated 6 February 2015, and all Federal, State and local guidelines as applicable to the work involved.

1.2. Additionally, all work being performed will ensure compliance with the applicable National Fire Protection Association national consensus standards listed below.

NFPA 54, National Fuel Gas Code
NFPA 58, Liquefied Petroleum Gas Code
NFPA 70, National Electrical Code®
NFPA 405, Standard for the Recurring Proficiency of Airport Fire Fighters
NFPA 1003, Standard for Airport Fire Fighter Professional Qualifications
NFPA 1402, Standard on Facilities for Fire Training and Associated Props
NFPA 1403, Standard on Live Fire Training Evolutions

1.3. Installation, testing and replacement of gas piping, gas utilization equipment or accessories, repair and servicing of equipment, shall be performed only by qualified personnel. In the interest of safety, all persons employed handling Liquefied Petroleum Gas (LPG) shall be trained in the proper handling and operating procedures which the employer shall document. After major maintenance or a major problem is discovered, the owner is obligated to contact the Air Force Civil Engineer Center (AFCEC), the commissioning authority to determine what actions will be taken. AFCEC will determine if the AFTF will remain out-of-service and determine if a site visit is necessary. Examples of major problems include: Any safety related condition that could lead to an imminent hazard such as: LPG piping or tank corrosion that has reduced the wall thickness to less than that required for the maximum allowable operating pressure. Any crack or other material defect that impairs the structural integrity or reliability of an LPG tank or piping. Any abnormal movement by environmental causes, such as earthquakes, landslides or floods that may impair the serviceability of a pipeline, tank, or controls. Any malfunction or operating error that

causes the pressure of an LPG pipeline or tank to rise above its maximum allowable working pressure or a leak in the LPG pipeline or tank that constitutes an emergency.

2. INSPECTION, TESTING AND MAINTENANCE SERVICES. The contractor shall maintain the AFTF by scheduling and conducting two semiannual inspections (not to exceed 180 days) in conjunction with winterization/de-winterization (Appendix B) activities where applicable. The contractor shall perform all inspections, testing and maintenance in accordance with (IAW) the current edition of TO 35E1-2-13-1. The inspections shall be IAW Table 1 Work Package (WP) 006 and Table 3 WP 007 of the TO. The contractor shall provide a written summary of the conditions found during the inspection. The contractor shall provide a cost estimate to repair/replace all items identified as discrepancies. The estimated labor cost to repair the AFTF, shall be quoted to the installation prior to conducting any repairs. Materials, supplies, and parts needed for AFTF maintenance and/or repairs, which are not readily available from government stock/supply may be purchased by the contractor if approved by the contracting officer or manager. Any item(s) purchased costing over (\$1000.00) shall be reimbursed to the contractor but must be approved by the contracting officer or manager prior to purchase.

2.0.1 Igniter Replacement. The contractor shall remove and replace the igniters in accordance with manufacture instructions, industry practices and procedures. There are 18 igniters that may require replacement, six are located on the ground in the burner area and 12 are located on the mock-up.

2.0.2 Cutout Panel Replacement. If the training scenario included forced entry extrication through cutout panels, the contractor shall install new cutout panels. The contractor shall remove and replace damaged, cutout panels in accordance with manufacturer's instructions and standard industry practices. Cutout panels are located on the left and right sides of the mock-up. The panels are 36 inches wide by 85.5 inches high and constructed of .036-inch thick sheet metal.

2.0.3 Water Pump Replacement. The contractor shall remove and replace the water pump in accordance with manufacturer's instruction and standard industry practices.

2.0.4 High Density Polyethylene (HDPE) Liner Maintenance. The 80-mil thick HDPE liners are located in the water conservation pond, burn area, and trench. The liners in the trench and burn pit are buried. The contractor shall locate the liner leak and repair in accordance with extrusion welder manufacturer's instructions. The contractor shall submit for the COR approval of the repair procedures, materials, and techniques prior to the specific repair of the HDPE liners.

2.0.4.1 Examine all seams and non-seams areas of the geo-membrane to identify holes, blisters, defects, unsupervised raw materials, and any signs of contamination by foreign matter. Submit repairs or corrective action requirement, in writing, to the COR within 48 hours.

2.0.4.2 Do not weld or replace geo-membrane during precipitation or if excessive moisture, blowing dust, high winds, or standing water is present.

2.0.5 Regulator Adjustment. Contractor shall adjust regulator pressure as follow:

<u>Burner Location</u>	<u>Optimal Flame Height</u>
Engine Burners	Fully Engulfed
Internal Burners	6-8 Feet

2.0.5.1 Storage Tank Maintenance and Repair. Ensure proper maintenance and repair of the 10,000 gallon propane storage tank, in accordance with specification and regulations provided.

2.0.5.2 Propane Supply. Contractor shall maintain the propane level at no less than 25% during use season, and must be filled to 80% capacity. The contractor will be reimbursed at cost for the propane and must submit receipts for reimbursement.

2.0.5.3 Quarterly Safety Inspections. As a minimum, the contractor shall conduct quarterly inspections on the following items.

2.0.5.4 Special Items. Expansion joints bolts on mock-up shall be inspected.

2.0.5.5 Inspection Items. The following items, as a minimum, shall be inspected:

2.0.5.6 Inspect burn area. Ensure rocks are level so when flooded a continuous film of water covers area from side to side. Ensure that rock level is ½ inch to 1 inch below top of burnarea drain.

2.0.5.7 Inspect burn area drain valve and wire drain valve for freedom of operation and absence of obstruction.

2.0.5.8 Inspect exterior and interior burners to ensure that burner grates are not excessively corroded or warped and that igniter protective covers are in place.

2.0.5.9 Inspect igniters to ensure electrode is clean and that the ceramic insulator is not cracked or damaged. Ensure igniter tips are free of carbonization.

2.0.5.10 Inspect aircraft mock-up for excessive corrosion and structural integrity.

2.0.5.11 Inspect mock-up doors to ensure freedom of operation and no excessive corrosion.

2.0.5.12 Inspect all lighting for proper operation. Ensure all area lighting and cab lightings are operable.

2.0.5.13 Inspect propane system to specifications, to include a minimum, tank, gauges, valves, piping, compress air system, manual safety system and regulators for corrosion, wear, or serviceable limits.

2.0.5.14 Inspect control stand to include electrical and mechanical components, piping, valves, and regulators. Check for corrosion, wear, or serviceable limits. Inspect markings on valves and switches for legibility. Ensure all pressure gauges are operable.

2.0.5.15 Inspect the condition of the ignition and ground wires of each circuit. Twice per year check for and annotate voltage on each wire to ensure 10,000 volts at the igniters.

2.0.5.16 Inspect water conservation pond area. Examine visible portion of liner for damage.

2.0.5.17 Open the washout, sprinkler, and engine water valves and then apply power to water conservation pond pump and test for adequate operation and water pressure. Close all valves when test is complete.

2.0.5.18 Charge compressed air system and examine for leaks.

2.0.5.19 Ensure compressed air system and emergency shut-off pneumatic burner valves are functioning properly at all locations. Verify emergency shutdown system operability, through a function test and propane burner-off. Ensure that all pneumatic burner valve boxes are clearly accessible and uncovered (i.e. free of rock cover).

2.0.5.20 Pressurize propane piping for one (1) hour and examine valves for leaks at burners with flammable gas detector.

2.0.5.21 Inspect cathodic protection system. Report the cathodic protection potential readings for the system along with voltmeter and readings of the rectifier. A written report of the findings shall be submitted to the COR within 48 hours after inspection complete.

2.0.5.22 Inspect driving area around burn area for ruts. Level all ruts before proceeding.

2.0.5.23 Winterize Facility. November of each year, winterize the facility. Purge propane back to angle valve at 10,000 gallon tank. Shut-off power and bleed compress air system. Open all water jacket ball valves at the base of the water jackets.

2.0.5.24 De-winterize Facility. March of each year, de-winterize the facility. As a minimum the contractor shall turn on power to compressed air system, check for leaks, and closed all water jacket ball valves at the base of the water jackets.

2.0.5.25 Provide a written report of findings. The report shall include a detailed cost estimate for any repairs required. The contractor shall provide a written report to the COR within 48 hours, detailing the inspection findings and a cost estimate of any repairs outside the normal maintenance that is required.

2.1. High-Density Polyethylene (HDPE) Liner Maintenance. The 80-mil thick HDPE liners are located in the water conservation pond, burn area, and trench. The liners in the trench and burn pit are buried. The contractor shall locate the liner leak and repair it IAW extrusion welder manufacturer's instructions (National Seal Company, 1245 Corporate Blvd., Suite 300, Aurora, IL 60504. Telephone (708) 898-1161 or (800) 323-3820. The contractor shall submit for Civil Engineer Quality Assurance (QA) personnel approval the repair procedures, materials, and techniques prior to the specific repair of the HDPE liner(s). Additionally, the contractor shall adhere to the following general guidelines.

2.1.1. Use yellow plastic warning tape, buried 12-inches below the surface, and indicates underground wiring, anodes, grounding cells, or reference electrodes. The contractor shall properly replace the warning tape prior to backfilling.

2.1.2. Examine all seams and non-seam areas of the geomembrane to identify holes, blisters, defects, unsupervised raw materials, and any signs of contamination by foreign matter. Record repair/corrective action.

2.1.3. Will not weld or replace the geomembrane during precipitation or if excessive moisture, blowing dust, high winds, or standing water are present.

3. Response Times. The contractor shall respond to propane leak service calls by providing immediate verbal assistance and physical assistance as soon as practical (as agreed to between the government and the contractor). After the leak has been stopped, the contractor shall complete the repairs by the next workday. Liner maintenance response shall be determined by the severity of the damage. For all other repairs, the contractor shall respond within 5 workdays.

4. Reports. The Contractor shall submit a complete report of all inspections, maintenance and servicing accomplished to the Records Management (RM) Point of Contact (POC) within ten (10) days of inspection. The reports shall include a record of conditions found, discrepancies needing repairs, repairs accomplished, recharging, and all replacement parts used. All records, files, documents, regardless of media (e.g., paper, electronic, etc.), as described in the PWS, that are the responsibility of the contractor are the property of the Government and shall remain so upon termination or completion of the contract. The contractor shall keep these items current and maintain and dispose of them in accordance with the requirements established in Air Force

Instruction (AFI) 33-322, Records Management Program, and the Air Force Records Disposition Schedule, found in the Air Force Records Information Management System (AFRIMS), which can be provided by the RM POC upon request. Records shall be turned over to the Government upon completion of the contract unless otherwise stated. All records are subject to the Freedom of Information and Privacy Act. *The Joint Base Andrews Records Management Office (301) 981- 8350* will provide assistance and training to the contractor to ensure compliance with Federal record keeping requirements.

5. Environmental Controls. The contractor shall be knowledgeable of and comply with all applicable Federal, State, and local laws, regulations, and requirements regarding environmental protection. The contractor shall ensure policies and procedures are established that protect the health and safety of employees and the community and minimize/eliminate the risk of environmental pollution.

6. Notification of Environmental Spills. If the contractor spills or releases any hazardous substance into the environment, the contractor shall immediately report the incident to the QA, base fire department, and base civil engineering. The contractor shall be liable for containment and environmental clean up of the spill or release of such substance.

7. Safety. Liquid propane used in the facility is potentially explosive. Personnel shall practice proper grounding procedures IAW the National Electrical Code (NEC). Personnel working with or near high voltage equipment should be familiar with modern methods of resuscitation. Personnel shall wear appropriate protective clothing and be aware of emergency procedures to perform emergency shutdown and start-up procedures.

8. SERVICE DELIVERY SUMMARY.

Performance Objective	PWS Para	Performance Threshold
Maintain and Repair the AFTF. Items requiring maintenance are timely and correctly repaired or replaced. Submit accurate and timely reports.	2	The AFTF is operational at all times to permit ARFF training when needed. Submit reports within 48 hours. No more than 1 customer complaint per service month
Quarterly Safety Inspection. All areas of the AFRTF are inspected to ensure safe operation. Submit accurate and timely reports.	2.0.5.3	The AFTF can be safely operated at all time to permit crash fire rescue training when needed. Submit reports within 48 hours. No more than 1 deficiency per quarter.

Response Times. Response to propane leaks and liner maintenance are timely.	3	Provide immediate verbal assistance and respond to propane leaks as soon as practical. Respond to liner maintenance as conditions dictate.
Environmental Controls. Contractor shall comply with all Federal, State, and local environmental requirements.	5	All Federal, State, and Local Environmental requirement met. No more than 1 deficiency per quarter.
Safety. Contractor shall take appropriate measures to protect personnel, property, and materials from chemical, electrical, and other mishaps.	7	Personnel, property, and materials are protected at all times.

9. GOVERNMENT FURNISHED PROPERTY AND SERVICES. The Government will provide electricity, water and sewer, utilities, police protection, fire protection, and emergency medical services. Any medical expenses incurred by contractor personnel shall be the responsibility of the contractor.

10. GENERAL INFORMATION.

10.1 PERIOD OF PERFORMANCE: The initial period of performance will be 27 September 2026 – 28 September 2027, and option years for this task may be exercised thereafter.

10.2. QUALITY CONTROL. The contractor shall develop and maintain a quality control program to ensure maintenance and repair services are performed IAW the TO and the manufacturer's instructions. The contractor shall develop and implement procedures to identify and prevent defective services from reoccurring. As a minimum, the contractor shall develop quality control procedures that address the areas identified in Paragraph 8, Service Delivery Summary. The government evaluator must have a specific quality control inspector to notify in case of customer complaints.

10.3. QUALITY ASSURANCE. The government will periodically evaluate the contractor's performance IAW the Quality Assurance Surveillance Plan.

10.3.1 Damages. The contractor shall repair or replace any Government property that is properly marked, and damaged by the contractor at no additional expense to the Government. For any work that may require digging, an AF Form 103 must be obtained and approved by 316 CES before any digging begins. Yellow plastic warning tape, buried 12 inches below the surface, indicates underground wiring, grounding cell, etc. The contractor shall properly replace the warning tape

prior to backfilling. The contractor will not be liable for damage to equipment that isn't properly marked.

10.4. GOVERNMENT REMEDIES. The contracting officer shall follow the requirements of FAR 52.212-4, *Contract Terms and Conditions for Commercial Items* (October 2018), for contractor's failure to correct nonconforming services.

The initial period of performance will be 15Sept2025-15Nov2025, and option years for this task may be exercised thereafter.

10.5. HOURS OF OPERATION. The contractor shall schedule with the Training and Logistics office 15 working days prior to arrival to coordinate escort and fire department training area gate access. Work hours shall be during normal duty hours (0700 to 1600), Monday thru Friday and weekends, excluding: New Year's day, Martin Luther King, Jr. Day, Presidents Day, Memorial Day, Independence Day, Columbus Day, Thanksgiving Day, Veterans Day, Christmas.

10.5.1 Emergencies. In the case of an emergency, the COR, Base Installation Fire Chief, base operations officer or their representative, shall have the authority to order the contractor to terminate work and clear the area of personnel and equipment.

10.6. SECURITY REQUIREMENTS. The contractor shall conform to all Base Regulations and Directives that pertain to security, safety, traffic, fire and personnel clearances. To obtain passes for all work on JBA, MD the contractor shall submit the following information on each employee before expected arrival on base after award of contract to the COR.

Full Name of
Individual Social
Security Number
Company Name
Vehicle Driver's License #
Vehicle State and Tag Number
Project individual id on (contract number and title)

Contract personnel shall then have to report to Visitor's Center to obtain passes.

10.6.1. A base identification card/vehicle pass shall be obtained for all employees including any subcontractor. It is the intent of the Government that each employee has possession of his pass at all times while working on this contract. The pass shall be displayed at all times below shoulders and above the waist. The contractor shall be responsible for the return of any pass when employment/and the contract is terminated.

10.6.2. All Government issued identification must be returned to the COR upon expiration of the contract or employee termination, whichever comes first. Failing to do so will result in the withholding of the final payment.

10.6.3. The Contractor shall be responsible for safeguarding all Government property provided for Contractor's use. At the end of each shift, all Government facilities, equipment, and materials shall be secured. The Contractor shall be responsible for providing and placing all barricades and safety devices during any activities and comply with all Occupational Safety and Health Standards (OSHA).

10.6.4. The contractor shall be responsible for the action of any of his subcontractor and his employees who are working on JBA, MD.

10.7. WEAPONS. Weapons of any sort are not authorized on JBA, this includes guns, knives, swords, numb chucks, brass knuckles, or any other weapon that has a sole purpose to injure or maim. Combination tools such as Leatherman and small pocket knives with folding blades less than four inches are authorized if used in the performance of duty. Those found in violation of this directive are subject to apprehension, criminal charges, and debarment from the installation.

10.8. ILLEGAL CONTROLLED SUBSTANCE. The U.S. Federal Government is the most vocal opponent of the drug industry and it has set the de facto international standards regarding the legality and illegality of different drugs. Illegal controlled substance and narcotics are not authorized on JBA what so ever, controlled substances may not be transported on the installation via vehicle or on an individual person, i.e., marijuana/Cannabis, cocaine, crack cocaine, LSD, Methamphetamine (Meth), Heroin, Anabolic steroids, Psilocybin mushrooms, are some of forbidden substances (this list is not all inclusive). Prescription medications are authorized for the person the medication is issued to; however the medication must not be expired and the label must clearly indicate the name of the person the medication is issued to. Otherwise the prescription medication may be deemed illegal by the law enforcement officials. Those found in violation of this directive are subject to apprehension, criminal charges and debarment from the installation.

10.9. PERFORMANCE OF SERVICES DURING A CRISIS DECLARED BY THE NATIONAL COMMAND AUTHORITY OR OVERSEAS COMBATANT COMMANDER. In the event the contractor is unable to perform his daily operation due to peacetime emergency and /or wartime contingency disturbances, etc., neither the Government nor the contractor shall take action for non-compliance with contract requirements.

Invoicing/payment for Reimbursable Items. The contractor shall invoice for

payment of reimbursable items. Each invoice must identify the chargeable contract line item. The billing shall be supported by an itemized list of all reimbursable parts and material used showing the items, unit cost, and total cost and copy of all invoices and sales slips. The contractor shall maintain original sales slips. The contractor shall maintain original delivery tickets, sales slips, or other documents identifying items purchased under the cost reimbursement portion of this contract. The COR shall accept and approve submitted invoices.

10.10. Ceremonial Events. IAW Air Force District of Washington (AFDW) I 90-1, The Contractor shall be required to observe quiet hours during outdoor ceremonial events. Quiet hours shall be in effect 30 minutes prior to commencement and one hour after the start of all outdoor ceremonial events.

11. APPENDICES.

- A.** Equipment Listing
- B.** System Particulars
- C.** Winterization/De-Winterization Procedure
- D.** Joint Base Andrews Environmental Protection Standards for Contracts

APPENDIX A

5.0 Appendix

A Equipment

Listing:

- AFTF Crash Trainer
- WRG Fire Training Simulation System - Tower Section #FC41LT & Residential Section #FCR3624 Structural Trainer
- WRG Fire Training Simulation System FCLA3000 & FCLA1000 Automated Live Fire Systems (4 total)
- WRG Fire Training Simulation Systems FCA206 Vehicle Fire Trainer
- WRG Fire Training Simulation Systems FCCST42 Confined Space Trainer
- WRG Fire Training Simulation Systems FCF/040 Flash Over Identification Trainer
- WRG Fire Training Simulation Systems FCHM306 Tractor Trailer Roll Over Hazmat Trainer
- Helo Fire Trainer
- Fire Fighter Challenge Stair Tower
- Storage Building
- Pump Room and Pumps
- Compressor Building and Compressor
- Propane Tank
- All Related Infrastructure and Parts

Appendix B

System Particulars

1. This theory of operation is based on the generic design. The Aircraft Fire Training Facility (AFTF) consists of an aircraft mock-up, burn area, control stand, liquid propane fuel system, cathodic protection system and a water conservation pond.
2. The aircraft mock-up, located in the burn area, is constructed of non-coated/unpainted carbon steel. It is approximately 72 feet long, 25 feet high at tip of tail, with a wingspan of 72 feet. The mock-up is configured with three simulated aircraft engines, two on the right wing and one located on the vertical stabilizer. Each engine contains one burner assembly and two ignitors. The wiring to the engine ignitors is enclosed in a protective water jacket that protects it from possible failure due to extreme heat. Water injection is located in both wing engines and at some sites in the tail engine. The wing engines also have dams installed to simulate a three-dimensional flowing fuel type fire. Wheel well/brake and auxiliary power unit fires are simulated on the right side of the fuselage adjacent to the inboard engine. In the interior of the mock-up, ignitors and burners are installed to simulate flight deck, passenger compartment, and baggage/cargo hold fires. There is also a standalone aircraft debris fire located away from the main fuselage of the aircraft. A water sprinkler system is installed in the top of the mock-up to cool and washout the mock-up. An expansion joint is located midway on the fuselage to allow for expansion and contraction during training. Cutout panels are located on the left and right aft sides of the fuselage. These 36-inch x 84-inch replaceable panels are constructed of .031-inch-thick sheet metal. The panels are used to simulate emergency entrance into an aircraft. Two steel hinged doors are located at the forward left and left amid ship.
3. The burn area is 100 feet in diameter. It is recessed and lined with one 80-mil thick HDPE liner. The liner is sandwiched between 6-inch-thick layers of sand and then covered by a layer of geotextile fabric and then an 18-inch layer of compacted aggregate crushed stone.
4. There are eight ignitors and burners installed in the burn area to simulate seven exterior ground fires which includes the debris fire and one running fuel fire with pool fire on the outboard engine. The ground fires are located on the left and right sides of the forward fuselage, between the right-wing engines, under the left-wing, and on the left and right sides of the aft fuselage. The weir is used to maintain the proper water level in the burn area. Opening the burn area drain valve allows all the water in the burn area to drain to the water conservation pond.
5. Located on the control stand are the valves, solenoids, and switches used to control the propane flow and ignitors during training events. Turning off the master/emergency switch located on the control stand will turn off power to all ignitors and close the internal valve located on the propane tank via the pneumatic control system, if applicable. The propane piping, water piping, and electrical conduits from the control stand to the burn area are located in an underground HDPE lined trench.

6. The liquid propane system supplies fuel for training exercises. Propane is supplied from a 10,000 gallon-WC (water capacity) tank to the various burners in the burn area through a series of pipes and valves.

7. A propane pump may be installed to obtain adequate pressure, if applicable. A corresponding valve located on the control stand, controls propane flow to each individual burner. Safety features of the propane system are provided to stop the flow of fuel to the burners, and will include a pneumatic or manual control system, and an emergency shut-off valve located on the control stand. Pressure relief valves are installed on the piping as required by the current edition of National Fire Protection Association Standard 58, *Liquefied Petroleum Gas Code*. A cathodic protection system is used to protect below ground piping from corrosion through sacrificial anodes inside the trench from the control stand to the burn area.

8. The water conservation pond is a 72 feet by 72 feet square area that is 9 feet deep. It is lined with an 80-mil thick HDPE liner. Unlike the burn area, the liner is not covered. A pump located in the pond is used to supply necessary water for the training area.

APPENDIX C

Winterization/De-Winterization Procedures

a. Winterization shutdown activities will consist of:

- ☐ Test all igniters and verify their proper operation, (identify repairs needed for the spring start up).
- ☐ Winterize each igniter by cleaning them with compressed air and applying non-freezing liquid to dissipate any moisture that may be collected by the igniter throughout the winter.
- ☐ Cover and securely seal each individual igniter.
- ☐ Verify that the rectifier for the impressed current cathodic protection system is operational.
- ☐ Lubricate and service all switches, cabinet hinges, latches, etc.
- ☐ Remove and store the water pond aerator as directed by the owner.
- ☐ Verify that the water conservation pond pump has a minimum of three feet of water for cover or remove it from the pond and store as directed by the owner.
- ☐ Disconnect the water-cooling pump in the burn area and store if applicable.
- ☐ Introduce a minimum of 90 Cubic Feet per Minute (CFM) of compressed air into the water system to blow out all water lines. Maintain air until all moisture is out of the lines to keep them from freezing during the winter.
- ☐ Exercise, service and lubricate all valves for the water system.
- ☐ Drain the burn area to an absolute minimum and install a temporary sump pump if practicable to prevent freezing of electric, propane and nitrogen systems.
- ☐ Verify that all automatic pumping systems are functional for the winter.
- ☐ Check all burner locations for performance and emergency shutdown, (identify repairs needed for the spring start up).
- ☐ Exercise the thermal cycles on the propane system and perform a soap test at all propane connections, (identify repairs needed for spring start up).

- ☐ Introduce a minimum of 90 CFM compressed air into the propane system and remove all water from the propane lines and coax valves. Maintain air until all moisture is out of lines to keep them from freezing during the winter.
- ☐ Pressurize the nitrogen/air system and perform a leak test on all nitrogen/air connections, (identify repairs needed for spring start up).
- ☐ Drain the nitrogen/air accumulator tank of all moisture.
- ☐ Service all nitrogen/air pilot valves.
- ☐ Provide a complete semiannual inspection report IAW TO 35E1-2-13-1.
- ☐ Provide a complete report of all work performed and minor repairs made.
- ☐ Brief and provide a written report to the Fire Chief or representative of all minor and major repairs required for the spring start up. Provide the user with a cost estimate for repairs to allow time for the using agency to secure funds by the spring start up.

b. De-winterization/spring start up activities will consist of:

- ☐ Verify that the rectifier is in service for the cathodic protection system.
- ☐ Lubricate all electrical panel hinges, switches, etc.
- ☐ De-winterize all igniters and assure good solid (functional) ignition.
- ☐ Check that any applicable pumps and/or lift stations are fully operational.
- ☐ Check and verify that all identification and warning labels are installed, clear and legible.
- ☐ Exercise the water system and adjust to assure that all functions are operating at the proper pressures.
- ☐ Fill the burn pit to the operating level and assure that the igniter coolers are adjusted properly.
- ☐ Install the water aerator (if applicable) in the pond.
- ☐ Treat the water in the water conservation pond to help prevent algae growth if necessary.

- ☐ Verify that the mechanical emergency shutdown valves associated with filling the propane tank are functioning properly and serviced.
- ☐ Check that the propane tank float gauge is operational.
- ☐ Check that the propane tank roto gauge is operational.
- ☐ Check that the propane tank pressure gauge is functioning.
- ☐ Perform maintenance per the manufacturer's recommendations on all propane valves and equipment.
- ☐ Operate the entire propane system assuring that all burners are functioning properly and that all emergency shutdown times are achieved.
- ☐ Cycle all propane system valves and perform leak tests at all connections.
- ☐ Perform common maintenance on all propane parts and equipment as recommended by the manufacturer.
- ☐ Perform a pressure test on the nitrogen/air system.
- ☐ Service and cycle all nitrogen/air pilot valves.
- ☐ Have a fire truck and crew stand by while testing live fire scenarios and making any burner adjustments, minor repairs, etc.
- ☐ Generate a complete report of all work performed and minor repairs made.
- ☐ Provide a complete semiannual inspection report IAW TO 35E1-2-13-1.
- ☐ Debrief and provide a written report to the Fire Chief or representative of all work performed and any findings.

APPENDIX D

JOINT BASE ANDREWS ENVIRONMENTAL PROTECTION STANDARDS for CONTRACTS

PART 1 GENERAL

1.1 SUMMARY

These standards address general environmental protection requirements for all construction, renovation, repair, and service contracts at Joint Base Andrews (JBA), MD. The U.S. Government, The U.S. Air Force, and (JBA), require that all contractors (to include all subcontractors) involved in the planning and execution of projects shall apply the general requirements identified herein to their specific projects and ensure compliance as applicable. This plan is not intended to be all-inclusive, but is to be used for all applicable contracts. It is the responsibility of the contractor to comply with all federal, state and local environmental laws and regulations in carrying out tasks associated with their project and it is the Air Force's responsibility to ensure that these standards are adhered to. This is a living document which is intended to be updated on a periodic basis as needed.

1.2 REFERENCES

All contractors and subcontractors shall comply with all applicable local, state (Code of Maryland Regulations), and federal regulations; all relevant Department of Defense and Air Force Instructions (AFI); and all (JBA) policies, plans, and programs. These include but are not limited to the following:

Regulation	Topic
40 CFR Parts 1500-1508	National Environmental Policy Act
32 CFR Part 989	Air Force Environmental Impact Analysis Process
40 CFR Parts 260-282	Hazardous Waste Management
49 CFR Parts 105-180	Transportation of Hazardous Material
COMAR Title 26 Subtitle 13	Hazardous Waste Management
AFMAN 32-7002; Chapter 5	Hazardous Waste Management
316 AW SPLAN 32-7042	Hazardous Waste Management Plan
AFMAN 32-7002; Chapter 3	Hazardous Materials Management
AAFBI 32-9003	Hazardous Materials Management
40 CFR Parts 61.115 & 763	Asbestos

AFI 32-1052	Facility Asbestos Management
COMAR Title 26 Subtitle 16	Lead
40 CFR Parts 112.1 – 112.3 & 300	Oil Pollution
COMAR Title 26 Subtitle 10	Oil Pollution and Tank Management
COMAR Title 26 Subtitle 10.01.12	Aboveground Storage Tanks
(JBA), Spill Prevention, Control, & Countermeasure Plan and Facility Response Plan	
Emergency Response	
40 CFR Part 761	Polychlorinated Biphenyls (PCBs),
	Manufacturing, Processing, Distribution in Commerce and Use
Prohibitions	
40 CFR Parts 60, 61, 63, 68 & 80	Air Quality
AFMAN 32-7002, Chapter 4	Air Quality
COMAR Title 26 Subtitle 11	Air Quality
40 CFR Parts 122 & 141	Water Quality
COMAR Title 26 Subtitle 08	Water Pollution
COMAR 26 Subtitle 17	Storm Water Management
(JBA) Storm Water Pollution Prevention Plan (SWPPP)	Storm Water Management
COMAR Title 26 Subtitle 17	Erosion and Sediment Control
2011 MD Standards and Specifications Soil Erosion & Sediment Control	Sediment Management
MD Erosion & Sediment Control Guidelines for State & Federal Projects 15 April 2010	Storm Water Management
JBA Storm Water Management Plan (SWMP)	Storm Water Management
Clean Water Act Section 404	Wetlands
AFI 32-7080	Pollution Prevention
(JBA) P2 Plan	Pollution Prevention
40 CFR Part 247	Affirmative Procurement
(JBA) Green Procurement Plan	Green Procurement
Executive Order 13101	Recycling and Acquisition
COMAR Title 26 Subtitle 04	Solid Waste
(JBA) ISWMP	Integrated Solid Waste Management Plan
29 CFR Part 1910.1200	Hazard Communication
40 CFR Part 230	Wetlands
COMAR Title 26 Subtitle 23	Non-tidal Wetlands
50 CFR Part 17	Endangered & Threatened Wildlife & Plants
COMAR Title 8 Subtitle 3	Threatened & Endangered Species, Wildlife Management (Lethal and Non-Lethal)
AFI 32-7064	Integrated Natural Resources Management

COMAR Title 8 Subtitle 19	Forest Conservation
Memorandum of Understanding, MD & DoD	Coastal Zone Management, Forest Conservation
36 CFR Part 800	Protection of Historic Properties
32 CFR Part 229	Protection of Archeological Resources
AFI 32-7065	Cultural Resources Management
Executive Order 11990	Protection of Wetlands
AFI 64-117, Section 2.10	Affirmative Procurement
Executive Order 13423	Strengthening Federal Environmental, Energy, and Transportation Management
JBA Environmental Management System	Management of Environmental Risk
Executive Order 13990	Protecting Public Health and the Environment and Restoring Science To Tackle the Climate Crisis
AFMAN 32-1067	Water and Fuel Systems

1.3 ENVIRONMENTAL PROTECTION PLANNING REQUIREMENTS

The basic contracting requirements are as follows: 1) Provide and maintain environmental protection as required, during the life of the contract, 2) Plan for and provide environmental protective measures to control pollution that develops during normal construction activities or project execution, 3) Plan for and provide environmental protective measures required to correct conditions that develop during the construction of permanent or temporary features associated with the project, and 4) Comply with federal, state and local regulations pertaining to the environment including, but not limited to, water, air and ground pollution.

1.3.1 Environmental Impact Analysis

The Environmental Impact Analysis Process (EIAP), Title 32 Code of Federal Regulations, Section 989 is the Air Force procedure for complying with the National Environmental Policy Act of 1969 (NEPA). The AF Form 813, Request for Environmental Impact Analysis, is used for documenting the need to conduct environmental analysis or the application of certain Categorical Exclusions (CATEX). The EIAP should address all known environmental constraints within and surrounding the construction project footprint, and shall be incorporated into contract specifications. Compliance with requirements identified in the EIAP documents is mandatory. A completed AF Form 813 must be retained by the Contracting Officer before a contract is awarded.

1.3.2 Environment Protection Planning

All contractors shall be required to demonstrate that they are capable of complying with all applicable environmental laws and regulations. It is the responsibility of each contractor to ensure environmental compliance requirements are properly identified and addressed in all meetings, site visits and included in all contract plans and submittals. In most instances a separate Environmental Compliance Plan, as identified by the 316 CES/CEI (Installation Management Flight) through the Contracting Officer (CO) or their representative, may be required to ensure full site awareness and thorough compliance. Environmental planning conducted by the contractor shall, at a minimum, address the following as applicable:

1.3.2.1 Laws, Regulations and Required Permits

The contractor shall prepare a submittal identifying all applicable federal, state and local laws, regulations and permit requirements concerning environmental protection, pollution control, abatement, and the requirements imposed by those laws, regulations and permits. All submittals shall be IAW specification 01300 Submittals. All requirements shall be routed concurrently through the CO(R) and 316 CES/CEI for review.

1.3.2.2 Protection of Environmental Features

The contractor shall determine methods for the protection of features to be preserved within the authorized work areas. The contractor shall submit methods to protect resources needing protection, including but not limited to: trees (i.e. tree protection fences), shrubs, grasses and ground cover; landscape features and soil; air and water quality; fish and wildlife; and historical, cultural and archeological resources.

1.3.2.3 Emergency Procedures

The contractor shall submit procedures to be followed as necessary in order to mitigate pollution of the environment from their construction site due to accident, natural causes or equipment failure.

1.3.2.4 Drawings

The contractor shall submit drawings showing locations of any proposed temporary excavations or embankments, haul roads, drainage systems, stream crossings, material storage areas, structures, parking areas, equipment storage areas, sanitary facilities, stockpiles of earth materials, disposal areas for excess earth material/unsatisfactory earth materials, and any other project-related temporary disturbance.

1.3.2.5 Environmental Monitoring and Sampling Plans

The contractor shall include environmental monitoring and sampling plans for all applicable soil, water and air compliance requirements.

1.3.2.6 Work Area Plan

The contractor shall submit a work area plan showing the proposed activity in each portion of the area and identifying the areas of limited use or non-use. The plan shall include measures for marking the limits of use areas. The proposed work areas shall not exceed those limitations or boundaries identified in the environmental document prepared for this project.

1.3.3 Environmental Permits

The contractor shall be responsible for obtaining all required environmental permits prior to the commencement of construction, if the necessary environmental permits are not provided. In addition, the contractor shall complete the necessary studies or documentation to obtain said permits. The contractor shall provide concurrently to the CO(R) and 316 CES/CEI, copies of all environmental permit applications and supporting documents before submission to any regulatory agency. Upon issuance, the contractor will provide the CO(R) with copies of the approved permit(s), the CO(R) will provide copies of all permits to the 316 CES/CEI.

1.3.4 Environmental Training

The contractor shall ensure that all of its personnel are properly trained, equipped and aware of all applicable environmental regulations and requirements. In addition, the contractor shall be accountable for ensuring all of its personnel comply with said requirements. The contractor shall maintain documentation of this environmental training and provide 316 CES/CEI (via the CO) access to those training records as required.

1.3.5 Environmental Submittals

The contractor shall identify environmental submittals required throughout the project life, including closeout; procedures for submittal to 316 CES/CEI through the CO(R); and an estimated schedule of environmental submittals.

PART 2 EXECUTION

2.1 HAZARDOUS MATERIALS

2.1.1 Hazardous Materials Use

The contractor shall provide a list of all hazardous materials to be used to the CO(R) and 316 CES/CEI, concurrently. IAW AAFBI 32-9003 a “Contractor Hazmat Notification Sheet” shall be filled out identifying types and quantities brought on base, used on base as part of a contract, and removed from base upon contract completion. In addition, all Safety Data Sheets (SDS) shall be provided to the CO and 316 CES/CEI. Compliance with 29 CFR§ 1910.1200, Hazard Communication Requirements, is mandatory at all times.

2.1.2 Hazardous Material Prohibitions

The use of Class I ozone depleting compounds, asbestos containing materials, polychlorinated biphenyls (PCBs), and mercury containing materials are prohibited. In addition, products containing EPA-17 compounds are discouraged. In all cases 316 CES/CEI, in conjunction with the CO and upon contractor request, may waive in writing any prohibited material IAW applicable laws and regulations.

2.1.3 Unforeseen Hazardous Materials

If a material that is not indicated on any contract documents or drawings is encountered, and determined to be potentially dangerous to human health upon disturbance during construction operations, the contractor shall stop that portion of work and notify the CO immediately. This would include polychlorinated biphenyls (PCB), lead paint, asbestos, contaminated soils, etc. The CO will determine the appropriate course of action.

2.1.4 Hazardous Waste

All hazardous waste generated by or as a result of contractor activities shall be managed, stored, transported and disposed of in accordance with all applicable federal, Maryland, and local regulations.

Hazardous waste storage on site shall be limited to initial accumulation point criteria. Each waste stream generated shall be limited to no more than 55 gallons of hazardous waste or 1 quart of acute hazardous waste on site at any one time. Establishment of hazardous waste storage sites shall be pre-coordinated via the CO with 316 CES/CEI Hazardous Waste Manager (301-981-2579).

All hazardous waste shall be properly classified and profiled by the contractor in order to ensure proper disposal.

Management of any hazardous waste generated as the result of contractor activities shall be the sole responsibility of the contractor. The contractor must ensure that any hazardous waste

generated through the contractor's activities is properly removed from the site and disposed of in accordance with applicable law. No hazardous waste shall be left at the site upon completion of contract operations.

All hazardous waste shall be properly manifested and transported IAW CFR Title

49. All hazardous waste manifests shall be coordinated with and signed by 316 CES/CEI through the CO, unless otherwise changed in writing by the CO. Fully executed copies of manifests will be returned to 316 CES/CEI for recordkeeping, in accordance with federal law.

2.1.5 Spills

The contractor shall take all necessary precautions to prevent spills of hazardous materials to include oils and hazardous wastes. In the event of a spill, the contractor shall immediately notify the fire department by calling 911, the CO, and 316 CES (301-981-5151). The contractor shall ensure that the applicable requirements of federal and state regulations as well as JBA spill provisions are complied with. The contractor shall have on hand the necessary resources (Absorbent pads, booms, spill absorbent materials, drain plugs) to properly respond to and clean up any spill caused by their operations. Any direction from the CO or the JBA incident commander concerning a spill or release that is required to protect human health or the environment shall be immediately and properly executed.

2.2 AIR QUALITY

2.2.1 Required Air Permits

Air quality permits to construct and/or permits to operate may be required for any stationary equipment or device that emits pollutants to the atmosphere or controls emissions to the atmosphere unless exempted by regulation. The 316 CES Air Quality Manager must review any permits for new construction or permits to operate.

Typical air construction permits include those for:

- Fuel burning boilers and heaters with a heat input rating of 1,000,000 Btu per hour or greater
- Stationary emergency generators with a rating of 373 kW (500 hp) or greater
- Motor vehicle gasoline storage tanks with a capacity greater than 2,000 gallons
- Paint spray and abrasive blasting booths
- Groundwater remediation systems that do not qualify for any of the de minimis exemptions in COMAR 26.11.02.10

- Non-residential char-broilers or pit barbecues with a cooking area greater than 5 square feet

Typical air operating permits include:

- Fuel burning equipment and turbines with a heat input capacity of 50 million Btu per hour or greater
- General incinerators with a rated capacity of 2,000 lbs per hour or greater and other specialty incinerators as identified by regulation
- VOC storage tanks with required vapor control
- Motor vehicle fuel storage tanks with a capacity of 40,000 gal or larger
- Gasoline tank truck loading racks where more than 20,000 gallons per day are loaded

Prior to the start of construction activities, the contractor shall prepare applications for any Air Permits required by the Maryland Department of the Environment (MDE).

All applications shall be submitted to the CO(R) for and the 316 CES/CEI, concurrently for review and approval, prior to submission to the MDE. The contractor shall ensure adequate time for the preparation and submission of permit applications to avoid any adverse impacts and/or delays to their project. The contractor shall be responsible for the payment of all applicable permit fees.

2.2.2 Emissions

Stationary internal combustion engines shall not exceed 10% visible emissions from idling engines and 40% visible emissions from operating engines. Exceptions are noted in COMAR 26.11.09.05 B (4). Fuel burning equipment shall not discharge visible emissions.

Boilers, furnaces, water heaters and other fuel burning equipment with a maximum heat input rating of 1 million Btu per hour or greater shall utilize low NOx technology. Equipment with a heat input rating of less than 1 million Btu per hour should use similar technology, if possible.

Contractor shall not conduct open burning at any sites on base.

Contractor shall respond appropriately to minimize dust emissions. Soils at the site and on unpaved access roads shall be treated with dust suppressants. Use of

dry power brooms and air blowing is not authorized. Wet cutting shall be used for all masonry type cutting such as concrete, concrete blocks, stone, etc. Contractor shall provide tarpaulin drop cloths and windscreens under and around sandblasting operations to confine and collect dust, sand, paint, and other debris for disposal.

Contractor shall not use or apply cutback asphalt in any paving operations, except in the situations noted in COMAR 26.11.11.02 (C).

Contractor shall use architectural coatings that comply with the Volatile Organic Compound (VOC) limits identified in COMAR 26.11.33, and coatings that comply with the miscellaneous metal, aerospace and vehicle refinishing limits identified in COMAR 26.11.19.

Contractor shall not vent or cause to be vented to the atmosphere chlorofluorocarbon (CFC) or hydrochlorofluorocarbon (HCFC) refrigerants, halon or other similar ozone depleting compounds. Contractor shall have available refrigerant recovery/recycling equipment and personnel certified to operate said equipment. All reclaimed refrigerant and halon shall be disposed of through an approved reclamation center or be properly reused on the project (in some cases specific CFCs and halons remain the property of the government and must be returned to the Defense Logistics Agency CFC/Halon Bank). Proof of proper disposal shall be provided to the CO and 316 CES/CEI.

Contractors performing work on facility Heating Ventilation and Air Conditioning (HVAC) equipment shall coordinate with the 316 CES/CEO HVAC Shop (through the CO) to provide required information on service performed and the amount and type of refrigerant disposed of.

2.3 WATER RESOURCES

2.3.1 Design for Storm water Management

For projects requiring post-construction storm water management, the contractor will, at a minimum comply with the current edition of Maryland Storm water Management Guidelines for State & Federal Projects and the Maryland Storm water Design Manual, Volumes I & II. The contractor may use other storm water management techniques as long as they comply with Maryland laws and regulations. To this end, the contractor is encouraged to integrate comprehensive environmental site design methods into the process when designing storm water controls.

The Contractor will make the use of non-structural Best Management Practices (BMPs) the first priority when designing storm water management controls. The contractor may use structural BMPs only after all practical non-structural options are exhausted. If the contractor bypasses non-structural BMPs for structural BMPs, the contractor will explain the reasoning for this choice in the design summary to the government. Exposed storm water management ponds, especially ones that contain standing water, are to be used only as a last resort and not at all within the confines of the airfield or flight paths.

The contractor will evaluate the impacts to the overall watershed resulting from the planned construction project and any storm water controls. The contractor will include in this evaluation the impacts of other planned projects in the same watershed that combined may cause synergistic storm water issues beyond the contractors' own project. If this evaluation uncovers consequences outside of the contractor's scope of work, the contractor will notify the CO of these issues in the design summary.

2.3.2 Existing Government Storm Water Resources

The contractor will refer, as applicable, to the current version of the following base plans when designing post-construction storm water management for projects:

- (JBA) General Plan
- (JBA) Storm Water Pollution Prevention Plan (SWPPP)
- (JBA) Storm Water Institutional Management Plan (IMP)
- (JBA) Wetlands Delineation
- (JBA) Floodplain Analysis
- Andrews Oil-Water Separator Management Plan
- (JBA) Spill Prevention, Control and Countermeasures (SPCC) Plan
- (JBA) Storm Water Management Plan (SWMP)

The contractor can find information on Maryland's storm water management requirements at the MDE web site, located online at:
<http://www.mde.state.md.us/Programs/WaterPrograms/SedimentandStormwater/index.asp>.

2.3.3 Storm Water Permits

The contractor shall prepare a National Pollutant Discharge Elimination System (NPDES) discharge permit application prior to the start of construction activities IAW Maryland State law and regulation, currently projects with a limit of disturbance greater than 1 acre. Presently, turnaround times are estimated at 6 weeks to 3 months; however turnaround times will vary depending on the scope of the project. The Contractor must retain the services of a MDE-approved permit expeditor in order to assist in obtaining this permit. The permit application shall be submitted to the CO(R) and the 316 CES/CEI concurrently prior to submission to the MDE. The Contractor shall obtain the approved permit from MDE and make all necessary notifications before construction start. A Storm Water Pollution Prevention Plan (SWPPP) must be developed and maintained in accordance with the permit. Weekly inspections of storm water management and sediment and erosion controls must be performed and documented. A copy of the permit, the SWPPP, inspection records and all other applicable documents shall be maintained at the job site or a centralized and accessible location.

In the event MDE issues a General Storm water Permit for Construction Activities, the Contractor will submit a Notice of Intent (NOI) for coverage under the General Permit for projects with a limit of disturbance greater than 1 acre. The NOI shall be submitted to the CO(R) and the 316 CES/CEI concurrently prior to submission to the MDE. The Contractor shall obtain the approved NOI from MDE and make all necessary notifications before construction start. A Storm Water Pollution Prevention Plan (SWPPP) must be developed and maintained in accordance with the general permit. Weekly inspections of storm water management and sediment and erosion controls must be performed and documented. A copy of the NOI, the Maryland General Storm water Permit for Construction Activities, the SWPPP, inspection records and all other applicable documents shall be maintained at the job site or a centralized and accessible location. Contractor will forward any MDE SWM inspections to 316 CES/CEI for recordkeeping and entry into EASIER.

Contractor will provide a disc copy of any Storm Water BMP As-builts to 316 CES/CENME (Org Box: 316.CES.Execution.Support@us.af.mil), JB Andrews Execution Support. Contractors are required to deliver these as-builts for all Storm Water BMPs implemented or updated. As-builts can and will be requested at any point in time if not delivered or missing. Contact 316 CES/CENME for physical address.

2.3.4 Storm Water Management Plans

The contractor shall prepare and submit a storm water management plan for all projects that exceed 5,000 square feet in size. The plans shall be coordinated with the 316 CES/CEI through the CO prior to submission to the MDE. The contractor shall comply with the conditions set forth in the plan/approval.

2.3.5 Sedimentation and Erosion Control Plans

The contractor shall prepare a sedimentation and erosion control plan for all projects that exceed 5,000 square feet or more than 100 cubic yards of soil. The plans shall be coordinated with the CO and 316 CES/CEI before submission to the MDE. The contractor shall comply with the conditions set forth in the plan/approval.

2.3.6 Sedimentation and Erosion Control Protection

The contractor shall implement and maintain all necessary controls to prevent sedimentation runoff to storm water for all construction projects regardless of size or scope. The contractor shall use the current version of Maryland's Erosion & Sediment Control Guidelines for State and Federal Projects and the Maryland Standards and Specifications for Soil Erosion and Sediment Control.

The contractor shall ensure a properly trained and certified individual (holder of a "MDE Green Card") is on site during applicable activities. Discharge of sediment to storm water (including storm drains) is prohibited.

The contractor shall execute soil stabilization to include seeding, hydro seeding, etc. within 7 days after soil disturbance on slopes and 14 days in other areas.

Contractor shall employ appropriate measures to prevent discharge of sediment to storm water during construction dewatering activities.

Excess materials generated from contractor activities to include soil, slurry, gravel, concrete or other similar materials must be removed from JBA. Disposal or dumping of these materials on base is not authorized. The only exception to this is if the materials are to be reused on the project that generated the excess material. Temporary storage is only authorized within the project area and must implement all required sedimentation and erosion control measures.

2.3.7 Borrow Material

No material shall be borrowed from on site locations without the approval of 316 CES/CEI.

Fill material from offbase shall be tested for the presence of petroleum constituents (TPH, BTEX), TCLP Metals, ignitability, corrosivity, and reactivity. It shall contain less than 5 parts per million (ppm) of TPH, less than 5 parts per billion (ppb) of benzene, less than 5,000 ppb of the sum of benzene, toluene, ethylbenzene, and xylenes, and shall not fail the TCLP test. TPH concentrations shall be determined by using EPA Methods 8015B (TPH-GRO) and 8015B (TPH-DRO). Benzene and BTEX concentrations shall be determined using EPA Method 8021, and the TCLP test shall be conducted using EPA Method 1311. The tests shall be conducted on a composite sample of the fill material, with at least one test being conducted for each borrow site from which the fill material was obtained. The fill material shall not be brought onsite until the tests have been reviewed and approved by 316 CES/CEI.

No excess borrow material shall be disposed of on site. The temporary storage of borrow material shall be accomplished within the project site implementing all required sedimentation & erosion control measures.

2.3.8 Releases to Waterways

Discharges of any petroleum, oil and lubricants, hazardous material, sewage or other regulated substances to waterways, including storm drains, ditches or similar contrivances are strictly prohibited. Contractor shall employ all necessary measures to preclude contamination of waterways from the construction site.

Contractor shall take necessary precautions to prevent damage to buried pipelines, sewage lines, tanks and other infrastructure. Damage to such infrastructure shall be immediately reported the base fire department, the CO and the 316 CES/CEI as appropriate.

2.3.9 Sanitary Sewer Discharges

Contractors shall comply with the conditions of Washington Suburban Sanitary Commission (WSSC) Discharge Authorization Permit 00001, 8 August 2020, issued to (JBA). Contractors may refer to the 2018 WSSC Plumbing and Gas Code, available online at:

<https://www.wsscwater.com/files/live/sites/wssc/files/PlumbersInfo/2018%20Code%20Doc%20-%20Final.pdf> for further guidance. At a minimum, only standard domestic sewage is authorized for discharge to the base sanitary sewer system. No industrial waste, hazardous waste, hazardous material or other regulated substances may be discharged to the base sanitary sewer system. Contractors must contact the 316 CES/CEIE prior to contract start in order to make alternate waste disposal arrangements.

2.3.10 Oil Water Separators

Installation of below ground oil water separators (OWS) is highly discouraged and shall be authorized only where no other engineering option exists. OWS installations shall typically be above ground and of a high performance design to adequately “treat” the effluent being discharged to below sanitary sewer discharge criteria (as defined by base sanitary sewer discharge permit).

Process OWS shall be connected to the sanitary sewer system. OWS connections to storm water are prohibited except where designed and approved to “treat” storm water discharges.

OWS installations shall comply with all applicable construction codes and federal and state regulations.

2.4 TOXIC MATERIALS

2.4.1 Asbestos

The use of asbestos containing materials (ACM) is prohibited.

Contractor shall comply with all applicable measures to ensure asbestos is not released to the environment.

Contractor shall ensure that all waste ACM is properly containerized, labeled, manifested and disposed of at an authorized site. All waste manifests shall be coordinated with the CO for 316 CES/CEI review and signature prior to shipping.

2.4.2 Lead Based Paint (LBP)

The use of paints containing lead is prohibited.

2.4.3 PCBs

The use of products containing PCBs is prohibited.

2.5 TANKS

2.5.1 Installation of underground storage tanks (USTs) is strongly discouraged and shall be authorized by 316 CES/CD only where safety or security provisions outweigh the associated environmental risks. Aboveground storage tanks (ASTs) shall be the preferred installation. All new non- temporary tank installations must be coordinated with 316 CES/CEI.

2.5.2 All AST installations shall comply with Title 40 CFR Part 112 requirements for secondary containment and release prevention.

2.5.3 All temporary tank installations shall be ASTs. Temporary ASTs (TASTs) used to support construction operations shall comply with UL-142 and applicable National Fire Protection Association (NFPA) criteria. TASTs shall be double walled with adequate secondary containment at the fill port and discharge port to collect spills. The Contractor is responsible for maintaining a supply of spill response materials to respond to small spills in the work area, including absorbent pads and the capability to prevent a spill from entering a nearby storm drain or waterway using a plug, dike, or boom. Upon completion and demobilization of the AST, the area will be subject to inspection by the CO. Any spills, no matter how small the amount, should immediately be reported to the CO.

2.5.4 Permanent ASTs (PASTs) shall comply with UL-2085, and applicable NFPA criteria. PASTs shall be double walled with adequate secondary containment at the fill port and discharge port to collect spills.

2.5.5 All approved UST installations shall comply with all Title 40 CFR Part 280. 40,41,44 requirements to include containment, leak detection equipment (automatic tank gauging), cathodic protection (as applicable), and monitoring systems. Tank systems shall also comply with COMAR 26.10.02 – 05, and 26.10.07, which details tank construction requirements and release detection and inspection requirements.

2.5.6 All tank installations shall be properly signed and labeled with contents, capacity, any applicable hazards (e.g., flammable) and restrictions (e.g., no smoking within 50 feet).

2.5.7 316 CES/CEIE must be notified of all changes in service of UST and AST. Out of service UST and AST must be pumped out immediately upon change in service, fills locked, piping drained and disconnected, vents left open and unobstructed, and tanks clearly marked as out of service in accordance with 40 CFR 112.1 and COMAR 26.10.10.01.

2.5.8 Out of service ASTs should be removed and disposed of in accordance with UFC 3-460-01-14-1. USTs may only be left out of service for one year before removal in accordance with COMAR 26.10.10.02. All removal plans must be approved by 316 CES/CEIE.

2.6 RECYCLING

Recycling is mandatory for all personnel that live on or work for or on behalf of (JBA). This includes all host and tenant organizations as well as all contractor personnel. The following requirements apply to all contractors:

2.6.1 The contractor shall recycle, reclaim or reuse all materials to avoid, to the greatest extent possible, disposing of waste generated as part of the project in landfills. The contractor, through the CO shall report quantity of all waste generated and quantity recycled or reused to 316 CES/CEI on a quarterly basis.

2.6.2 Construction debris shall be recycled or reused to the maximum extent possible. Roll-off containers for the collection of ferrous and non-ferrous metals can be requested by calling the base Recycling Center at (301) 981-0644/0645

2.6.3 Residential and office-type recyclable materials such as white paper, cardboard, metal cans, glass bottles, plastic bottles, etc. shall be recycled, to the greatest extent possible. Contractors can call the base Recycling Center at 301-981-0644/0645 to request recycling bins for placement in offices and other areas of potential accumulation. Contractors can arrange for recyclables to be picked up or drop them off at the base Recycling Center, located at 3350 Celmer Lane (East side of base just North of the Pearl Harbor gate).

2.6.4 Green waste, such, as trees, limbs and other vegetation shall be processed, to the greatest extent possible, for composting.

2.6.5 Industrial type wastes such as tires, motor oil, batteries, etc., generated as a result of contractor project operations shall be recycled.

2.6.6 The contractor shall procure and use products with recycled content to the greatest extent possible. In addition environmentally preferred alternative products shall be given consideration to the greatest extent possible.

2.6.7. Green Procurement is the government's program designed to give preference to the purchase of materials with recycled content and/or bio- based content to include services that use such products. Comprehensive Procurement Guidelines are published by the EPA for those items that must legally be considered when making purchases. The Recommended Material Advisory Notices give guidance on specific percentages of recycled content in those materials listed by the EPA, for example the 30% requirement for copy paper. These documents can be found at <http://www.epa.gov/cpg/backgrnd.htm>.

Green procurement requirements apply to all federal agencies, as well as any state agency or contractor using appropriated federal funds. Written justification must be provided to purchase items that do not meet the minimum recovered material standards, using one of the following exemptions:

- The item is not available within a reasonable timeframe
- The performance of the recycled content item is inadequate
- The item is unavailable at a sufficient level of competition

The price of the recycled content item is unreasonable

Please refer to AFI 64-117, Section 2.10, for more information regarding affirmative procurement.

Examples of readily available products with recycled content include commercial carpeting, rugs, partitions, and other building systems.

2.7 CONTAMINATED MATERIALS (SOIL and GROUNDWATER)

2.7.1 Prior to any digging or trenching, the contractor shall comply with all conditions identified, as part of the approval. Digging or trenching within 6 feet of an existing groundwater monitoring well is prohibited. Contractors shall be liable to any damages to monitoring wells resulting from their activities. Should a well need to be abandoned; the work must follow Code of Maryland (COMAR) Title 26, subtitle 04 (26.04.04.11) for well "Abandonment Standards". This regulation requires an abandonment report which will have to be generated and signed by a Maryland certified well driller and provided to 316 CES/CEI.

2.7.2 Any contaminated soils that are encountered during construction activities shall be immediately reported to the CO and 316 CES/CEI (301-981-7121).

2.7.3 Construction projects that occur in locations where soil or groundwater contamination is known to exist or identified during construction activities shall comply with all applicable environmental and health and safety provisions. Contractor shall coordinate via the CO with 316 CES/CEI prior to construction activities for a site visit by a MDE inspector. In addition the following provisions shall apply:

2.7.3.1 A written Site Specific Health and Safety Plan (HASP) shall be prepared by the contractor IAW CFR Title 29, approved by a Certified Industrial Hygienist and coordinated with 316 CES/CEI through the CO prior to commencement of any construction activities.

2.7.3.2 All personnel working in the construction area shall be familiar with all aspects of the Site Specific HASP. Personnel shall be trained in hazardous waste operations to the extent required by 29 CFR Part 1910.120.

2.7.3.3 During excavation, soils shall be monitored with a photo ionization detector for evidence of contamination. When PID readings above 10 ppm are encountered continuous screening of the excavated soil is required. Soils with PID readings of less than 50 ppm (taken from headspace in a bag) are considered “clean” and can be reused as backfill for the excavation, if suitable. Excavated soil, which exhibits signs of visual impact, shall be considered “contaminated.” Also, soils with PID readings of 50 ppm or greater, shall be considered “potentially contaminated.” If contamination or potential contamination is observed, the material shall be segregated from non-contaminated soils. The material shall be sampled for characterization before off-site disposal at a licensed waste management facility. Sampling and testing of excavated contaminated soils shall be conducted as specified in Section 2.7.3.5.

2.7.3.4 Excavated soil shall be placed in temporary storage (protected stockpiles or covered roll-off box) immediately after excavation. Stockpiles shall be constructed to isolate contaminated material from the environment including a geo-membrane liner on the ground surface, as well as a liner covering the material to prevent precipitation from entering the stockpile. The ground surface on which the liner is placed shall be free of rocks greater than 0.5 inches in diameter and any other object, which could damage the membrane. The cover material shall be anchored to prevent it from being removed by wind.

Samples of stored material shall be collected at a frequency of once per 25 cubic yards or per each roll-off container, whichever is less. Samples shall be tested for TPH using EPA Methods 8015B (TPH-GRO) and 8015B (TPH-DRO). If the stockpile is known or suspected to have been

impacted by/contained hydrocarbons heavier than No. 6 Fuel Oil (e.g., motor oil, lubricants, grease), then in addition to the listed methods, it shall also be tested for TPH using EPA Method 1664M. Stored materials with a TPH concentration that exceeds 50 ppm shall be considered contaminated and shall be treated offsite.

2.7.3.5 Contaminated soil shall be sampled prior to disposal IAW disposal site criteria. Prior to transportation the contaminated soil shall be tested for the following:

- Ignitability
- Percent Moisture
- pH Level
- VOCs (8260)
- PCBs (8080)
- TPH (8015 Modified – Purge & Trap)
- TPH (8015 Modified – Solvent Extraction)
- TCLP Metals (1311 for 8 RCRA Metals)
- Reactivity
- TPH (1664M)*
- TOX/Total Organic Halides (9020)*

* = These analyses are only necessary when characterizing impacts from waste oil or fuels heavier than No. 6 Fuel Oil (e.g., motor oil, lubricants, grease)

All collected analysis data shall be provided via the CO to 316 CES/CEI.

2.7.3.6 Contaminated liquids collected from excavations and stockpiles shall be temporarily stored in appropriate and approved containers. Liquid storage containers shall be watertight and stored to prevent any environmental risk from leaks or spills.

2.7.3.7 Contaminated liquid collected from excavations or stockpiles shall be sampled to the extent necessary to determine appropriate disposal methods IAW federal, state and disposal site regulations. Documentation of all analyses performed shall be furnished to the CO, who will forward to 316 CES/CEI and the treatment facility.

2.7.3.8 Contaminated materials from excavations or stockpiles shall be transported for off base disposal IAW federal and state regulations. All waste manifests shall be coordinated through the CO with 316 CES/CEI for signature prior to shipping, with fully executed copies provided upon disposal.

2.7.3.9 Excavations shall be backfilled with clean material immediately after all contaminated materials have been removed and the satisfactory results of confirmation tests for backfill material, specified in Section 2.3.5, have been approved by 316 CES/CEI.

2.7.3.10 The contractor shall prepare a letter and submit it via the CO to 316 CES/CEI within 45 calendar days of completing work at the site. The report shall be labeled with the contract number, project name, location, date and name of the general contractor completing the work. The letter shall include the following information as a minimum:

- Type of work performed
- Field screening readings
- Quantity of materials removed
- Sampling locations and sampling methods
- Sample chain-of-custody forms
- Source of backfill
- Chemical and physical test results
- Disposal manifests
- Certifications of final disposal

Drawings to show limits of excavation and contamination and sample locations
Photographs to document the soil removal, sampling procedures,
Contaminated material storage, fill placement and grading, and site post-construction

2.8 NATURAL RESOURCES

2.8.1 Any work on base by the contractor may not trench, dredge, drain or create crossings through any wetlands or within the 25-foot (100 feet in some cases) wetlands buffer without appropriate Clean Water Act sections 404 and 401 permitting by regional regulatory agencies and proper clearance via the CO from 316 CES/CEI. Wetland areas must be protected at all times by using proper Soil/Erosion Control Measures (see section) or other US Army Corps of Engineers or state approved methods. JBA has mapped wetland locations on base, and Air Force policy regarding wetland management is derived from compliance with Executive Order 11990, Protection of Wetlands, and is detailed in Air Force Manual (AFMAN) 32-7003, Chapter 3, Natural Resources Management. If projects overlap wetland areas, contact the CO and 316 CES/CEI.

2.8.2 The CO's approval is required before any equipment will be permitted to ford live streams. In areas where frequent crossings are required, install temporary culverts or bridges. Remove temporary culverts or bridges upon completion of work, and repair the area to its original condition or as indicated.

2.8.3 The contractor shall not disturb fish and wildlife without coordinating with 316 CES/CEIE Natural Resources Management. The contractor shall not alter water flows or otherwise significantly disturb the native habitat adjacent to the project, except as indicated or specified in writing. Disturbance includes, but is not limited to:

- Sealing building entrances ("exclusionary methods").
- Trapping, killing, removing, or injuring wildlife.
- Tree removal.

2.8.4 The contractor shall not impact plant or animal species without prior approval from 316 CES/CEI via the CO. Prohibited actions include impacting federal or state listed species, removing trees, harming birds, mammals or other animals.

2.8.5 The contractor shall comply with MDE stream closure periods.

2.9 HISTORICAL AND ARCHAEOLOGICAL RESOURCES

2.9.1 The contractor shall not conduct activities in any areas identified as having sensitive cultural resources without prior approval from 316 CES/CEI via the CO.

2.9.2 The contractor shall perform no work on base historic properties without prior approval from 316 CES/CEI via the CO.

2.9.3 The contractor shall carefully protect in-place and report immediately through the CO to 316 CES/CEI any historical or archaeological items or human skeletal remains discovered in the course of work. Contractor shall stop work in the immediate the area of the discovery until directed by the CO to resume work.

2.10 TREE REMOVAL

2.10.1 The contractor shall minimize the removal of trees to only those absolutely necessary to accomplish the scope of work. The project proponent must obtain an approved AFF 813 from 316 CES/CEI prior to the commencement of work. No tree clearing should occur from 1 March to 1 September.

2.10.2 If trees are to be removed, then they must be replaced by project proponent according to the following:

2.10.2.1 For removal of canopy cover of less than one (1) acre, one (1) tree shall be planted for each tree removed, according to a 1:1 ratio.

2.10.2.2 For removal of canopy cover exceeding one (1) acre, 60% of the canopy cover must be reforested.

2.10.3 Replacement trees must be native species, 2-5 inch caliper, replaced prior to tree removal (when possible), and designed IAW JBA's Arbor Plan (2011).

2.10.4 Replacement tree planting locations must be approved by 316 CES/CEIE Natural Resources Management and Environmental Planning prior to replanting.

2.10.5 Tree replacements must be maintained and monitored for a minimum of one year. A one year warranty for dying or dead trees shall be provided by the Contractor.

2.11 ENVIRONMENTAL MANAGEMENT SYSTEM

2.11.1 The contractor shall perform work consistent with the (JBA) environmental management system (EMS), to include all policies, procedures, objectives, and awareness training requirements.

2.11.2 The contractor shall perform work in a manner that conforms to all appropriate Environmental Management Programs and Operational Controls identified by the JBA EMS, and provide monitoring and measurement information as necessary for Andrews to address environmental performance relative to the environmental, energy, and transportation management goals.

2.11.3 In the event an environmental nonconformance or noncompliance associated with the contracted services is identified, the contractor shall take corrective and/or preventative actions. In the case of a noncompliance, the contractor shall respond and take corrective action immediately. In the case of a nonconformance, the contractor shall respond and take corrective action based on the time schedule established by the EMS Site Coordinator.

2.11.4 All contractor personnel present on-site for a period of six months or longer shall receive initial environmental EMS awareness training. Annual refresher training should also be provided. Supplemental, focused EMS training may also be required for some contractor personnel in order to meet established goals, objectives, and/or targets of related environmental management programs. Awareness training is available from 316 CES/CEI on disc or via the (JBA) intranet website. Additional training may be coordinated on an as needed basis.

2.11.5 The contractor shall ensure that their employees are aware of their roles and responsibilities as identified or implied by the EMS and their actions related to the performance of their work can potentially impact the environment. Information about the EMS and its requirements is available from 316 CES/CEI (301-981-3162) and will be provided to the contractor and/or CO when requested.

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